

METAL-CLAD PCBs To Dissipate Heat Generated In Modern Devices



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Insulated metal substrates take care of heat dissipation created by modern printed circuit designs

Light emitting diodes (LEDs) have been around for years. Nowadays, these are a major source of light. Unlike filament lamps, incandescent lamps, fluorescent tube-lights or compact fluorescent lamps, LEDs consume less power and have a prolonged life. Due to the availability of a wide range of LED wattages, these are used in residential buildings, in addition to industrial and flash/torch lights, for mobiles and medical electronics, and more, where high lumens are required. Today, technological advancements in LED design and processes are continually boosting light output.

However, LEDs generate a lot of heat, which needs to be managed for optimal operation and prolonged life. Power LEDs of greater than 1W are almost always surface-mounted devices. This is because axial leads to the die in a leaded package do not conduct enough heat away from the LEDs. Chip-on-board (COB), ceramic sub-mounts and other thermally-efficient packages are emerging as the standard thermal management packaging solution for power LEDs.

The effect of temperature

The LEDs' colour, or wavelength, changes with temperature. As die temperature increases, wavelength of the colour increases. This is particularly important with white light. The human eye can differentiate small colour changes in white light. When power LEDs are populated in an array, consistent thermal resistance from one die to the next assures consistent colour. Because of comparatively low thermal resistance that thermal clad offers in comparison to FR-4 (glass epoxy), die temperature is less affected by slight variances in junction-to-case thermal resistance that occurs with eutectic or epoxy-die mounting techniques.

It is also possible to pack the die more closely in an assembly that utilises good thermal management techniques, thereby reducing the effects of temperature. Generally, a 30 to 50 per cent drop in light output for a constant-forward current indicates end-of-life for power LEDs. Their lifetime has been extrapolated to be more than 50,000 hours.

Insulated metal substrates (IMSeS) take care of heat dissipation created by modern printed circuit designs. IMS materials offer an effective alternative to other solutions, such as heat-sinks and ceramic substrates, which are often linked to interconnection difficulties.

Care must be taken while selecting IMS for an application. In general, IMSeS minimise thermal impedance and conduct heat more effectively. These substrates are mechanically robust than thick-film ceramics and direct-bond copper constructions that are often used in these applications.

Aluminium-based copper-clad laminates, the most common form of IMS, are used widely in LED lighting circuits. However, there are many other applications in fields such as power supplies, voltage regulators, hybrid power ICs, amplifiers, ignition controls and relays, in industries as diverse as automotive, medical equipment, communications and audio.

Standard construction of IMS

IMS is a unique layered system comprising the following layers:

Copper foil. There are two types of copper foils: electrodeposited (ED) and high-temperature elongation (HTE). ED copper foils are produced by depositing copper on a stainless-steel drum that rotates in a plating tank with anodes of very pure copper. Thickness of the copper foil is determined by the length of the plating process, that is, rotation speed of the drum.

Copper is then removed from the drum and wound on a roll. The finished copper foil is very smooth on the drum side and fairly rough on the other. The rough side has the advantage of giving better adhesion to the base laminate.

HTE foils are manufactured by annealing ED copper at a high temperature. Because of higher ductility, such foils are less prone to

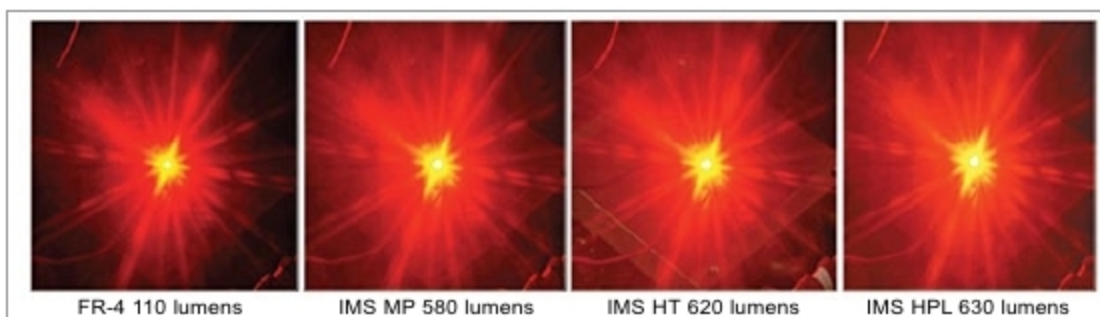


Fig. 1: Light output of the same LED die on different circuit board materials at a maintained die temperature of 80°C (Credit: Bergquist)